Lecture 5 Normal walking kinematics
SM2501/FSG3084

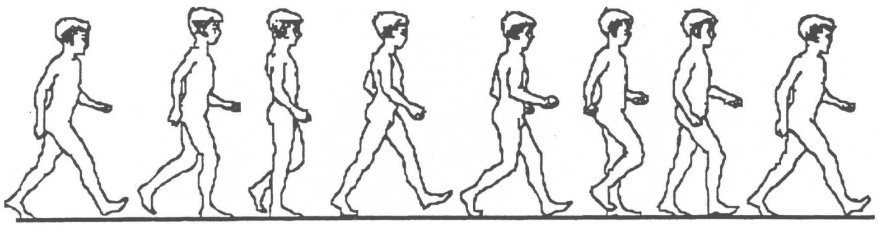


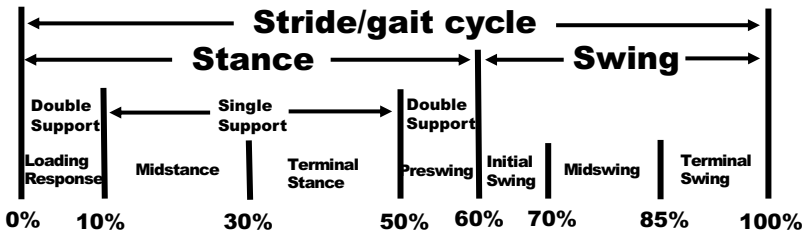
Abbey Road, Beatles,1969

Objectives:
Introduce terminology of walking/gait analysis
Describe segment and joint kinematics in gait

1

Gait Cycle Terminology



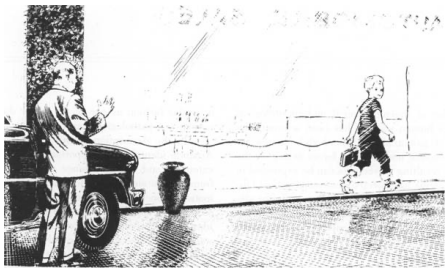


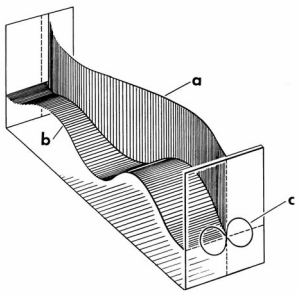
2

THE MAJOR DETERMINANTS IN NORMAL AND PATHOLOGICAL GAIT *†

BY J. B. DEC. M. SAUNDERS, M.B., F.R.C.S.(EDIN.), VERNE T. INMAN, M.D., AND HOWARD D. EBERHART, M.S., SAN FRANCISCO, CALIFORNIA

From the Departments of Anatomy and Orthopaedic Surgery, School of Medicine, and The College of Engineering, University of California, San Francisco



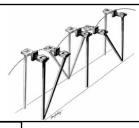


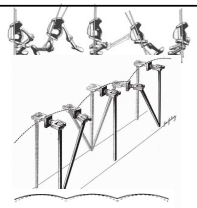
- Center of mass (COM) oscillates
- Fast and large motions are less energy efficient

3

Review: "The major determinants in normal and pathological gait", Saunders et al., 1953

Premise: "compass gait": COM trajectory as arcs with sharp transitions -> inefficient





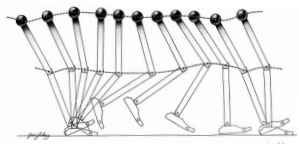
Pelvic rotation



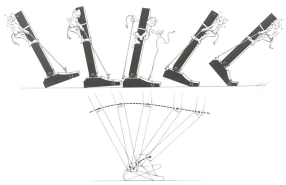
Pelvic elevation



Knee flexion in stance



Knee and hip flexion interaction



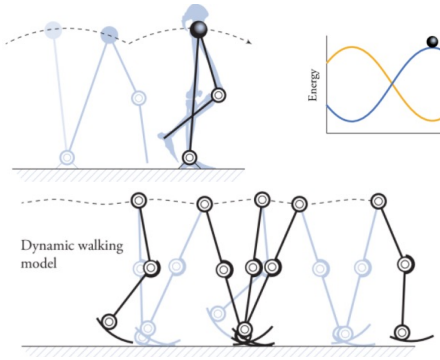
Heel rise, knee and ankle interaction

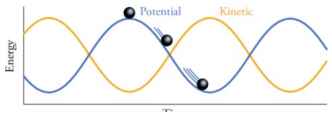


Lateral pelvis displacement; hip adduction + knee valgus

4

Since 1953: normal gait w.r.t. mechanical energy efficiency





Delp & Uchida 2020

> J Biomech. 2014 Apr 11;47(6):1324-31. doi: 10.1016/j.jbiomech.2014.02.002. Epub 2014 Feb 14.

Quantitative evaluation of the major determinants of human gait

Yi-Chung Lin [1], Margit Gfoehler [2], Marcus

> Gerontology. 2022;68(2):151-161. doi: 10.1159/000515326. Epub 2021 May 10.

Effects of Short- and Long-Term Aerobic-Strength Training and Determinants of Walking Speed in the Elderly

Lucia Slobodová [1 2], Ľudmila Oreská [3], Martin Schön [1 2], Patrik Krumpolec [1 4], Veronika Tirpáková [3], Peter Jurina [3], Jakub Laurovič [3], Matej Vajda [3], Michal Nemec [1], Eva Hečková [4], Ivana Šoóšová [2], Ján Cvečka [3], Dušan Hamar [3], Peter Turčáni [4], Chia-Liang Tsai [4], Wolfgang Bogner [4], Milan Sedliak [3], Martin Kršiak [4 5], Jozef Ukropec [1], Barbara Ukropcová [1 2]

Review > Hum Mov Sci. 2007 Aug;26(4):617-56. doi: Epub 2007 Jul 6.

The six determinants of gait and the inverted pendulum analogy: A dynamic walking perspective

Arthur D Kuo [1]

Dynamic principles of gait and their clinical implications

> Gait Posture. 2007 Jul;26(2):295-300. doi: 10.1016/j.gaitpost.2006.09.079. Epub 2006 Nov 21.

Determinants of gait as applied to children with cerebral palsy

S D Russell [1], B C Bennett, D C Kerrigan, M F Abel

> Arch Phys Med Rehabil. 2001 Feb;82(2):217-20. doi: 10.1053/apmr.2001.18063.

Quantification of pelvic rotation as a determinant of gait

D C Kerrigan [1], P O Riley, J L Lelas, U Della Croce

> Gait Posture. 2001 Oct;14(2):79-84. doi: 10.1016/s0966-6362(01)0

A refined view of the determinants of gait

U Della Croce [1], P O Riley, J L Lelas, D C Kerrigan

5

Energy cost of walking

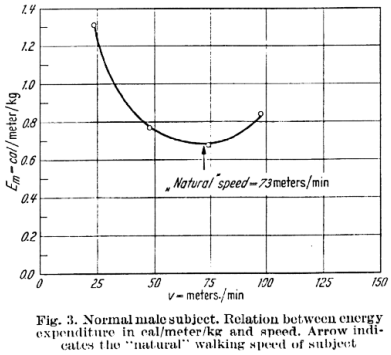
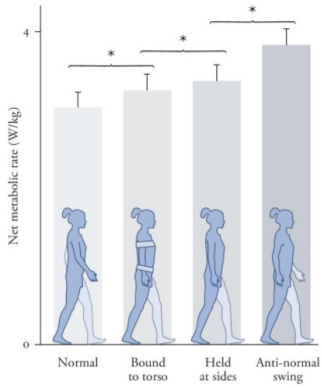


Fig. 3. Normal male subject. Relation between energy expenditure in cal/meter/kg and speed. Arrow indicates the "natural" walking speed of subject

Ralston 1958



Delp & Uchida 2020

worksheet

6

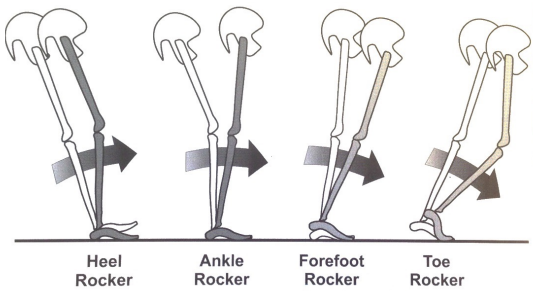
"Prerequisites of Normal Gait" (J Gage 1991)
underlying features of normal walking

- Stability in stance
- Foot clearance in swing
- Pre-positioning of foot for initial contact
- Adequate step length
- Energy conservation

To polygon

7

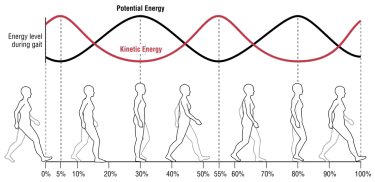
Stance phase as foot and ankle "rockers" (J Perry 2010)



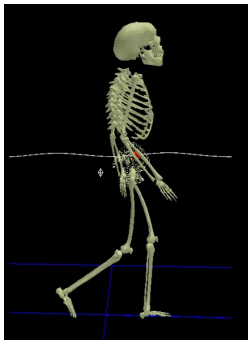
Foot rockers, Jacquelin Perry, 2010

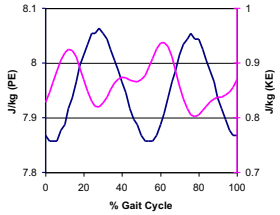
8

"65% recovery of mechanical energy through pendular motion" Cavagna et al. 1976, Dipaola et al. 2016, Bennett et al, 2005



Neumann 2002

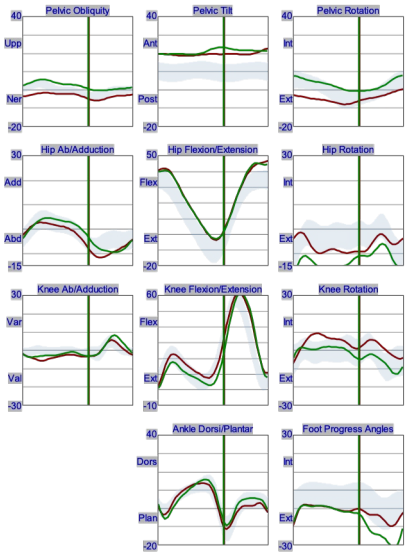




Experimental observation

9

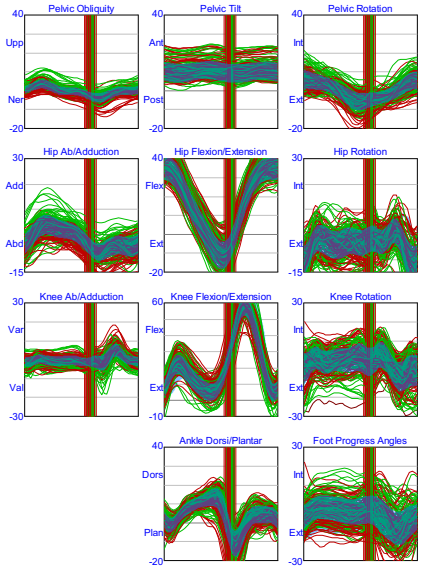
Is this person 'abnormal'



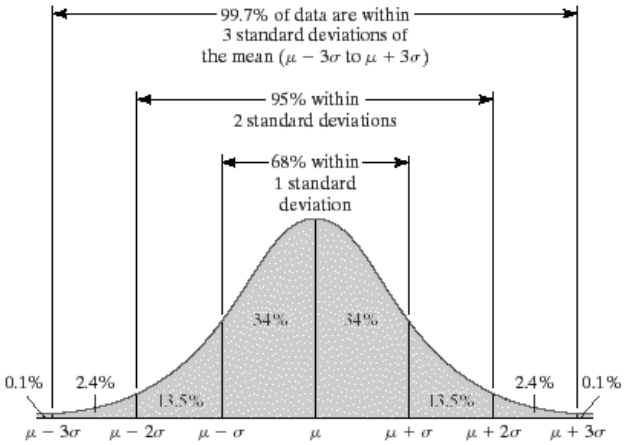
10

'Normal' database

- Mean ± 1 SD includes how much normal data?



11



Normal distribution

12